

TEMIR, Kazakhstan

WMO: 354260

Lat: 49.1422N

Lon: 57.1217E

Elev: 768

StdP: 14.29

Time Zone: 5.00 (E05)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-18.7	-13.4	-24.8	1.4	-18.2	-19.6	1.9	-13.1	20.5	11.3	18.3	12.7	3.3	320	0.538

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	22.9	96.8	65.4	93.7	65.0	90.6	64.2	68.3	88.4	67.1	87.2	65.9	85.6	7.5	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
62.3	86.4	73.4	60.5	81.0	72.9	58.7	75.7	73.3	33.0	89.2	31.9	86.8	31.0	85.9	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.3	15.8	13.7	DB	-24.0	101.6	7.1	3.3	-29.1	104.0	-33.3	105.9	-37.3	107.8	-42.4	110.2	(4)
(5)			WB	-24.0	70.7	8.0	2.1	-29.8	72.2	-34.4	73.4	-38.9	74.6	-44.7	76.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	43.6	9.3	11.1	25.6	47.7	60.9	70.5	75.2	72.5	59.3	44.1	28.4	16.3
	DBStd	25.47	11.90	12.38	11.16	10.43	8.19	7.89	6.61	7.94	8.28	8.81	10.61	11.34
	HDD50	5221	1262	1089	758	164	11	1	0	0	17	226	649	1044
	HDD65	8686	1727	1509	1220	522	179	35	6	27	205	648	1099	1509
	CDD50	2885	0	0	2	97	349	615	782	699	297	43	1	0
	CDD65	874	0	0	0	4	53	199	322	261	35	0	0	0
	CDH74	11180	0	0	0	77	728	2596	3968	3253	546	12	0	0
	CDH80	5259	0	0	0	7	237	1223	1965	1628	198	1	0	0
Wind	WSAvg	5.9	6.1	6.2	6.8	6.6	6.2	5.8	5.6	5.5	5.2	5.5	5.6	5.9
Precipitation	PrecAvg	10.7	0.9	0.8	0.7	1.1	1.0	0.9	1.0	0.7	0.7	0.9	1.1	1.0
	PrecMax	13.9	2.8	2.8	2.2	3.3	3.1	3.0	3.2	3.4	3.1	2.8	2.3	2.8
	PrecMin	6.2	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.1	0.0
	PrecStd	2.1	0.6	0.6	0.5	0.8	0.8	0.7	0.8	0.6	0.6	0.6	0.6	0.7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	33.5	34.9	60.1	80.9	90.0	98.9	100.8	100.5	94.0	74.9	54.1	38.9
		MCWB	32.1	32.7	46.6	59.0	62.4	66.3	66.3	65.5	62.4	54.2	47.9	34.7
	2%	DB	31.7	32.8	50.9	75.9	86.1	95.1	96.6	96.2	85.1	69.5	49.1	34.6
		MCWB	30.6	31.5	43.5	57.3	61.8	65.4	65.5	65.1	58.7	52.7	44.2	32.5
	5%	DB	29.3	30.8	43.3	70.7	82.1	91.5	93.2	92.9	80.1	64.5	44.8	32.8
		MCWB	28.2	29.5	38.4	54.2	61.0	64.6	65.1	64.8	57.1	49.7	40.9	31.2
	10%	DB	26.5	28.3	38.5	65.6	77.4	87.5	90.1	89.1	75.4	59.4	41.1	30.6
		MCWB	25.4	27.0	35.4	51.4	58.8	63.1	64.6	63.4	56.0	47.6	38.2	29.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	32.4	33.2	48.7	61.0	67.3	71.5	70.1	69.3	64.0	57.4	50.7	35.6
		MCDB	33.2	33.9	57.2	76.8	84.8	92.2	88.9	90.5	89.1	70.1	53.6	37.6
	2%	WB	30.6	31.7	43.4	57.8	64.6	68.0	68.4	67.4	60.4	53.9	44.6	33.2
		MCDB	31.6	32.7	50.3	73.2	80.2	87.3	88.1	89.6	79.9	66.0	47.8	34.4
	5%	WB	28.2	29.6	38.9	55.2	62.0	66.2	67.2	66.0	58.6	51.2	41.3	31.4
		MCDB	29.2	30.7	43.2	68.8	77.1	86.0	86.8	88.1	75.8	61.7	44.5	32.7
	10%	WB	25.4	27.0	35.5	52.4	59.8	64.5	66.2	64.6	56.9	48.5	38.3	29.1
		MCDB	26.4	28.2	38.2	64.2	74.2	83.2	85.4	85.2	73.2	58.3	40.9	30.5
Mean Daily Temperature Range	5% DB	MDBR	13.4	14.7	14.6	20.1	22.4	23.9	22.9	24.1	24.1	20.3	12.4	12.1
		MCDBR	10.4	10.9	18.7	27.4	28.9	28.5	27.0	28.1	31.2	29.0	16.8	11.2
		MCWBR	9.8	10.1	12.4	13.3	11.2	9.9	8.4	9.9	13.0	15.8	11.9	9.6
	5% WB	MCDBR	10.0	9.8	17.4	25.4	25.5	25.3	23.6	25.5	26.7	25.2	14.6	10.1
		MCWBR	10.0	9.5	12.1	13.1	11.1	10.2	8.4	9.5	12.2	15.2	11.3	9.2
Clear-Sky Solar Irradiance	taub	0.303	0.329	0.392	0.436	0.422	0.417	0.412	0.422	0.370	0.357	0.337	0.308	
	taud	2.084	2.020	1.985	2.048	2.172	2.248	2.275	2.220	2.319	2.277	2.210	2.114	
	Ebn at Noon	226	249	252	256	266	268	267	258	261	241	214	205	
	Edh at Noon	28	37	47	49	45	42	40	41	33	29	25	23	
All-Sky Solar Radiation	RadAvg	461	840	1185	1519	1940	2149	2081	1812	1349	770	368	295	
	RadStd	63	66	146	146	157	172	152	114	113	88	100	45	59

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (4 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	+1.98	N/A	N/A	N/A	N/A	+189	+151

Nomenclature: See separate page